

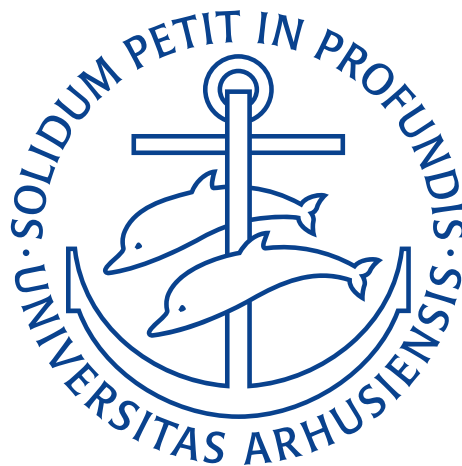
Take-Home Assignment week 37–38

Statik & Styrkelære

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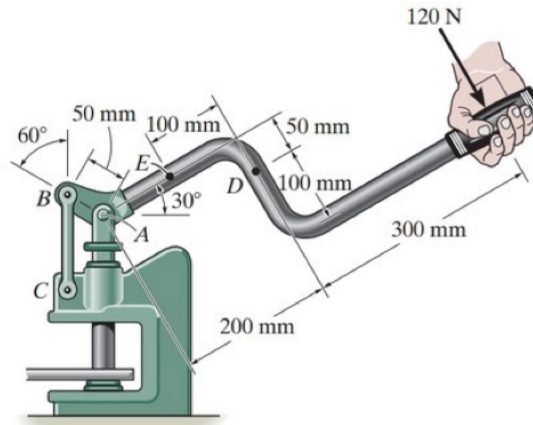
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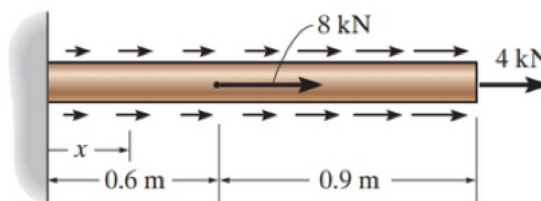
Mekanik & Teknologi
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Exercise P10-1

Determine the resultant internal loadings acting on the cross section at point E of the handle arm, and on the cross section of the short link BC .

**Exercise P10-2**

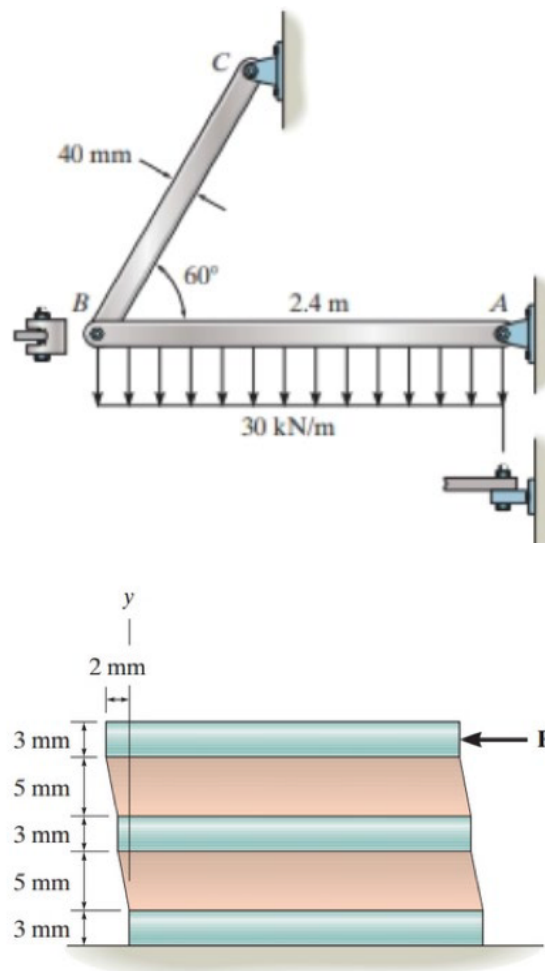
The bar has a cross-sectional area of $400 \cdot 10^{-6} \text{ m}^2$. If it is subjected to a triangular axial distributed loading along its length which is 0 at $x = 0$ and 9 kN m^{-1} at $x = 1,5 \text{ m}$, and to two concentrated loads as shown, determine the average normal stress in the bar as a function of x for $0 \leq x \leq 0,6 \text{ m}$.

**Exercise P11-1**

Determine the required thickness of member BC to the nearest mm, and the diameter of the pins at A and B if the allowable normal stress for member BC is $\sigma_{\text{allow}} = 200 \text{ MPa}$ and the allowable shear stress for the pins is $\tau_{\text{allow}} = 70 \text{ MPa}$.

Exercise P11-2

Nylon strips are fused to glass plates. When moderately heated the nylon will become soft while the glass stays approximately rigid. Determine the average shear strain in the nylon due to the load P when the assembly deforms as indicated.



Exercise P12-1

The rigid beam is supported by the three suspender bars. Bars AB and EF are made of aluminum and bar CD is made of steel. If each bar has a cross-sectional area of 450 mm^2 , determine the maximum value of P if the allowable stress is $(\sigma_{\text{allow}})_{\text{st}} = 200 \text{ MPa}$ for the steel and $(\sigma_{\text{allow}})_{\text{al}} = 150 \text{ MPa}$ for the aluminum. $E_{\text{st}} = 200 \text{ GPa}$ and $E_{\text{al}} = 70 \text{ GPa}$.

